

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### Product Identifier

Perihalan Produk: Ethanol, 70% v/v  
 Product Description: Ethanol, 70% v/v  
 Cat No. : TS/0484/25

### Relevant identified uses of the substance or mixture and uses advised against

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

### Company

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

### Supplier

E-mail address Enquiry.my@thermofisher.com

### Emergency Telephone Number

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

## SECTION 2: HAZARDS IDENTIFICATION

### Classification of the substance or mixture

|                                   |                   |
|-----------------------------------|-------------------|
| Flammable liquids                 | Category 2 (H225) |
| Serious Eye Damage/Eye Irritation | Category 2 (H319) |

### Label Elements



Signal Word

Danger

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## Hazard Statements

H225 - Highly flammable liquid and vapor  
H319 - Causes serious eye irritation

## Precautionary Statements

### Prevention

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P337 + P313 - If eye irritation persists: Get medical advice/attention  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

### Storage

P403 + P235 - Store in a well-ventilated place. Keep cool

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component     | CAS No    | Weight % |
|---------------|-----------|----------|
| Ethyl alcohol | 64-17-5   | 60-70    |
| Water         | 7732-18-5 | 30-40    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

If symptoms persist, call a physician.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

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Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

## Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### **Suitable Extinguishing Media**

Water mist may be used to cool closed containers.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### Special hazards arising from the substance or mixture

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental precautions

Do not flush into surface water or sanitary sewer system.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

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## Conditions for Safe Storage, Including any Incompatibilities

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component     | Malaysia | ACGIH TLV      | OSHA PEL                                                                                                         |
|---------------|----------|----------------|------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol |          | STEL: 1000 ppm | (Vacated) TWA: 1000 ppm<br>(Vacated) TWA: 1900 mg/m <sup>3</sup><br>TWA: 1000 ppm<br>TWA: 1900 mg/m <sup>3</sup> |

| Component     | European Union | The United Kingdom                                                                                            | Germany                                           |
|---------------|----------------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Ethyl alcohol |                | TWA: 1000 ppm TWA; 1920 mg/m <sup>3</sup><br>TWA<br>WEL - STEL: 3000 ppm STEL;<br>5760 mg/m <sup>3</sup> STEL | 200 ppm TWA MAK; 380 mg/m <sup>3</sup><br>TWA MAK |

### Exposure Controls

#### Engineering Measures

Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

#### Recommended Filter type:

Organic gases and vapours filter Type A Brown conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

### Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

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## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

#### Appearance

|                |                          |
|----------------|--------------------------|
| Physical State | Liquid                   |
| Odor           | No information available |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                             |                    |
|---------------------|-----------------------------|--------------------|
| Melting Point/Range | No data available           |                    |
| Softening Point     | No data available           |                    |
| Boiling Point/Range | 85 - 90 °C / 185 - 194 °F   | Estimated          |
| Flash Point         | 21 - 22 °C / 69.8 - 71.6 °F | Method - Estimated |

|                          |                   |        |
|--------------------------|-------------------|--------|
| Evaporation Rate         | No data available |        |
| Flammability (solid,gas) | Not applicable    | Liquid |
| Explosion Limits         | No data available |        |

|                              |                          |             |
|------------------------------|--------------------------|-------------|
| Vapor Pressure               | No data available        |             |
| Vapor Density                | No data available        | (Air = 1.0) |
| Specific Gravity / Density   | 0.85 - 0.88              |             |
| Bulk Density                 | Not applicable           | Liquid      |
| Water Solubility             | Miscible                 |             |
| Solubility in other solvents | No information available |             |

### Partition Coefficient (n-octanol/water)

|               |         |
|---------------|---------|
| Component     | log Pow |
| Ethyl alcohol | -0.32   |

|                           |                          |                                             |
|---------------------------|--------------------------|---------------------------------------------|
| Autoignition Temperature  | No data available        |                                             |
| Decomposition Temperature | No data available        |                                             |
| Viscosity                 | No data available        |                                             |
| Explosive Properties      |                          | Vapors may form explosive mixtures with air |
| Oxidizing Properties      | No information available |                                             |

|                |    |
|----------------|----|
| VOC Content(%) | 65 |
|----------------|----|

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
|--------------------------|---------------------------|

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## Hazardous Reactions

None under normal processing.

## Conditions to Avoid

Keep away from open flames, hot surfaces and sources of ignition.

## Incompatible Materials

Strong oxidizing agents. Strong acids. Acid anhydrides. Acid chlorides.

## Hazardous Decomposition Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral

Based on available data, the classification criteria are not met

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component     | LD50 Oral                                                    | LD50 Dermal | LC50 Inhalation                                                   |
|---------------|--------------------------------------------------------------|-------------|-------------------------------------------------------------------|
| Ethyl alcohol | LD50 = 10470 mg/kg<br>OECD 401 (Rat)<br>3450 mg/kg ( Mouse ) | -           | LC50 = 117-125 mg/l (4h)<br>OECD 403 (rat)<br>20000 ppm/10H (rat) |
| Water         | -                                                            | -           | -                                                                 |

##### (b) skin corrosion/irritation;

No data available

##### (c) serious eye damage/irritation;

Category 2

##### (d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                          | Test method                                                | Test species | Study result    |
|------------------------------------|------------------------------------------------------------|--------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 60-70 ) | Mouse Ear Swelling Test (MEST)                             | mouse        | non-sensitising |
|                                    | -----<br>OECD Test Guideline 429<br>Local Lymph Node Assay | mouse        | non-sensitising |

##### (e) germ cell mutagenicity;

No data available

| Component                          | Test method                                            | Test species         | Study result |
|------------------------------------|--------------------------------------------------------|----------------------|--------------|
| Ethyl alcohol<br>64-17-5 ( 60-70 ) | AMES test<br>OECD Test Guideline 471                   | in vitro<br>Bacteria | negative     |
|                                    | -----<br>Gene cell mutation<br>OECD Test Guideline 476 | in vitro             | negative     |

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|  |  |           |  |
|--|--|-----------|--|
|  |  | Mammalian |  |
|--|--|-----------|--|

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

| Component                          | Test method             | Test species / Duration      | Study result          |
|------------------------------------|-------------------------|------------------------------|-----------------------|
| Ethyl alcohol<br>64-17-5 ( 60-70 ) | OECD Test Guideline 416 | Oral / mouse<br>2 Generation | NOAEL = 13.8 g/kg/day |
|                                    | OECD Test Guideline 414 | Inhalation / Rat             | NOAEC =<br>16000 ppm  |

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

Ecotoxicity effects Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants. Do not empty into drains.

| Component     | Freshwater Fish                                                  | Water Flea                                    | Freshwater Algae                              | Microtox                                                                                                                |
|---------------|------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Ethyl alcohol | Fathead minnow<br>(Pimephales promelas)<br>LC50 = 14200 mg/l/96h | EC50 = 9268 mg/L/48h<br>EC50 = 10800 mg/L/24h | EC50 (72h) = 275 mg/l<br>(Chlorella vulgaris) | Photobacterium<br>phosphoreum: EC50 =<br>34634 mg/L/30 min<br>Photobacterium<br>phosphoreum: EC50 =<br>35470 mg/L/5 min |

### Persistence and degradability

**Persistence** Persistence is unlikely, based on information available.

| Component                          | Degradability   |
|------------------------------------|-----------------|
| Ethyl alcohol<br>64-17-5 ( 60-70 ) | OECD 301E = 94% |

Bioaccumulative potential Bioaccumulation is unlikely

| Component     | log Pow | Bioconcentration factor (BCF) |
|---------------|---------|-------------------------------|
| Ethyl alcohol | -0.32   | No data available             |

Mobility in soil The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility. Disperses rapidly in air.

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**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous Keep product and empty container away from heat and sources of ignition

**Other Information** Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Can be landfilled or incinerated, when in compliance with local regulations

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1170  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ETHANOL SOLUTION

### Road and Rail Transport

UN-No UN1170  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ETHANOL SOLUTION

### IATA

UN-No UN1170  
Hazard Class 3  
Packing Group II  
Proper Shipping Name ETHANOL SOLUTION

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

**International Inventories** China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS)

| Component     | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|---------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Ethyl alcohol | 200-578-6 | X    | X   | X     | X    | X    | X     | X    | KE-13217 |



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|       |           |   |   |   |   |  |   |   |          |
|-------|-----------|---|---|---|---|--|---|---|----------|
| Water | 231-791-2 | X | X | X | X |  | X | X | KE-35400 |
|-------|-----------|---|---|---|---|--|---|---|----------|

| Component     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|---------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Ethyl alcohol |                                                                                           |                                                                                          |                            | Annex I - Y42                      |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

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Revision Summary

Initial Release.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**